MILITARY SPECIFICATION

ALPHA-NITRONAPHTHALENE
(For Use in Explosives)

This specification has been approved by the
Bureau of Naval Weapons, Department of the Navy.

1. SCOPE

1.1 This specification covers one grade of alpha-nitronaphthalene commercially known as the technical grade.

2. APPLICABLE DOCUMENTS

2.1 The following documents, of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein.

STANDARDS

Military

MIL-STD-105	Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-129	Marking for Shipment and Storage
MIL-STD-286	Propellants: Sampling, Inspection and Testing

(When requesting any of the applicable documents, refer to both title and number. All requests should be made via the cognizant Government inspector. Copies of this specification and other unclassified specifications and drawings required by contractors in connection with specific procurement functions should be obtained upon application to the Commanding Officer, Naval Supply Depot (Code CDI), 5801 Tabor Avenue, Philadelphia 20, Pennsylvania. All other documents should be obtained from the procuring activity or as directed by the contracting officer.)

3. REQUIREMENTS

3.1 Material - The alpha-nitronaphthalene shall be a product of high quality, suitable for the purpose intended, and so manufactured as to meet the requirements specified herein.

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3.2 Melting point - The melting point of the alpha-nitronaphthalene shall be not less than 55.5°C when tested in accordance with 4.4.2.

3.3 Specific gravity - The specific gravity of the alpha-nitronaphthalene shall be 1.33 ± 0.03 grams per cubic centimeter when tested in accordance with 4.4.3.

3.4 Compatibility - The alpha-nitronaphthalene shall be compatible with 25/75 Cyclotol E explosive. When tested as specified in 4.4.4, the alpha-nitronaphthalene shall be considered compatible with the explosive when the net increase in gas evolution due to the alpha-nitronaphthalene-explosive combination shall not exceed 4.0 milliliters per 2 grams of test sample.

3.5 Workmanship - The alpha-nitronaphthalene shall be processed in a manner that will yield the high quality product necessary to meet the requirements of this specification (See 4.4.1).

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection - Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the supplier may utilize his own facilities or any commercial laboratory acceptable to the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.2 Acceptance inspection - Conformance of the alpha-nitronaphthalene to the requirements of this specification shall be determined entirely by means of acceptance inspection. The acceptance inspection shall consist of an examination for acceptability of quality control methods used by the manufacturer, examining and testing the acceptance samples (4.3.2) for all of the requirements of this specification, and an examination of the sample of filled containers (4.4.3) for conformance to the packaging, packing and marking requirements.

4.3 Sampling -

4.3.1 Size of lot - For the purpose of sampling, a lot of alpha-nitronaphthalene shall consist of a manufacturer's batch. If the material cannot be identified by batch, a lot shall consist of not more than 5,000 pounds of alpha-nitronaphthalene offered for delivery at one time.

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4.3.2 Sample for tests - From each lot offered for acceptance under contract, two 1-pound samples of alpha-nitronaphthalene shall be removed from separate unit containers taken at random. If more than one batch is included in the lot, two random 1-pound samples shall be taken from each batch.

4.3.3 Sample for examination of filled containers - A random sample of filled containers shall be selected from each lot of alpha-nitronaphthalene offered for acceptance under contract, in accordance with Standard MIL-STD-105 at inspection level I and acceptable quality level (AQL) = 2.5 percent defective.

4.4 Inspection methods -

4.4.1 Conformance of the alpha-nitronaphthalene to the requirements of 3.1 through 3.5 shall be determined by the following examinations and tests.

4.4.2 Melting point -

4.4.2.1 Apparatus - A melting point bath, equipped with a mechanical stirrer and a source of heat that can be easily regulated, shall be used. (A beaker of 1 to 2 liter capacity, approximately 3/4 full of clear paraffin oil, is suitable). An accurately standardized total-immersion Centigrade thermometer shall be suspended in the bath so that the bulb is not less than 1-1/2 inches from the bottom of the bath. If the mercury column of the thermometer is not completely immersed at the observed melting point temperature, a second thermometer shall be suspended approximately 1/2 inch from the first thermometer, with its bulb approximately at the height of the middle of the exposed mercury column of the first thermometer, for determining the correction factor.

4.4.2.2 Procedure - A thin-walled capillary tube of uniform diameter, long enough to extend beyond the top of the bath shall be used. The tube shall be filled with a dry portion of the sample to a depth of approximately 4 millimeters (mm), the sample compacted by tapping, and the tube fastened to the standardized thermometer. The stirrer shall be started and the bath heated to 40°C, then gradually heated so that the rise in temperature is not less than 1 degree in 3 minutes nor more than 1 degree in 1 minute. Melting point for this test shall be considered to be the temperature at which first indication of liquification is noted. If the mercury column of the thermometer is completely immersed at the observed melting point temperature, this temperature shall be reported, with thermometer calibration corrections applied, as the melting point of the sample. If part of the column is exposed, the following correction shall be added

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to the observed temperature:

$$\text{Correction} = 0.000159n\ (T-t)$$

Where:

n = Number of degrees in the exposed mercury column, in °C.

T = Uncorrected melting point, in °C.

t = Average temperature of the exposed mercury column, in °C.

4.4.3 Specific gravity - The test method used to determine the specific gravity shall be in accordance with the test method 510.1.1 (pycnometer method) of MIL-STD-286.

4.4.4 Compatibility test -

4.4.4.1 Sample preparation - Three (3) test samples, as enumberated below, shall be prepared. Each of the test samples shall then be separately tested in accordance with the vacuum stability test procedure of 4.4.4.2.

(a) A sample consisting of one (1) gram of 25/75 Cyclotol E explosive

(b) A sample consisting of one (1) gram of alpha-nitro-napthalene

(c) A sample consisting of an intimate mixture of one (1) gram of 25/75 Cyclotol E explosive, and one (1) gram of alpha-nitronaphthalene

4.4.4.2 Vacuum stability test -

4.4.4.2.1 The vacuum stability test is carried out in a glass unit, Figure 1. These tubes are calibrated before use and contain a volume of approximately 10 ml. One to two mm capillary bore Pyrex tubing is used for the manometer. The vacuum stability test chamber is a constant temperature block that accurately maintains a test temperature of 100° ± 0.5°C.

4.4.4.2.2 Calibration of glass tube - Determine the volume in ml of the 15.5 cm heating tube by running in mercury from a buret until the tube is filled to the level at which the ground glass joint of the capillary tube will make contact with mercury. Subtract from the indicated buret reading,

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the volume of explosive used in the test. The difference shall be represented by the symbol A. Transfer 7.0 ml of mercury to the cup at thw lower end of the capillary tube. Clmap the tube in an upright vertical position, and measure the height in mm of the mercury column in the capillary tube (approximately 25 mm.) Measure the length in mm of each of the 3 parts of the capillary tube and add these values to obtain total length. From the total length subtract the height of the mercury column in the capillary tube as previously obtained. Represent this difference by the symbol B_1. From the total length subtract the height of the column of mercury in the capillary tube measured at the end of the test described in 4.4.4.2.3. Represent this difference by the symbol B. Determine the capacity of the capillary tube per unit of length as follows: Transfer an accurately weighed sample of approximately 10 gm. of mercury to the cup at the lower end of the capillary tube. Manipulate the tube so that when it is horizontal, mercury is contained in a continuous section of the longest part of the tube and measure the length of the mercury column. Repeat this twice with the mercury in 2 other parts of the long section of the tube. Calculate the average of the 3 measured lengths of the mercury column. Represent the unit capacity in ml per mm of the capillary tubing by the symbol C. This can be obtained from the formula:

$$C = \frac{W}{DL}$$

C = Unit capacity of capillary tubing in ml per mm
W = Grams of mercury
D = Density of mercury at temperature of determination
L = Average measured lengths of mercury column in mm

4.4.4.2.3 Procedure - Transfer the sample to the heating tube of the apparatus shown in Figure 1. Connect the capillary tube to the heating tube. Clamp the apparatus so that the long section of the capillary tube is in a nearly vertical position. Transfer 7.0 ml of mercury to the cup at the lower end of the capillary tube. Connect a vacuum pump to the lower end of the capillary tube and evacuate the system until the pressure is reduced to approximately 5 mm of mercury. (Evacuation of the capillary tube is facilitated by placing the cup of the tube in a horizontal position so that mercury does not block the capillary opening.) After evacuation, disconnect the pump. Seal the connection between the capillary tube and the heating tube with 1 ml of mercury. Measure the total vertical height of the column of mercury in the capillary tube. Measure and subtract the vertical height of the mercury in the cup. The difference shall be represented by the symbol H_1. Note the room temperature (t_1) and the barometric pressure. Subtract the value H_1

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from the barometric pressure in mm. Represent this difference by the symbol P_1. Insert the heating tube in a constant temperature bath. Maintain at a temperature of 100.0°C ± 0.5°C for 48 hours. Remove the heating tube from the bath and allow to cool to room temperature. Measure the total vertical height of the column of mercury in the capillary tube and subtract the vertical height of the mercury in the cup. This difference shall be represented by the symbol H. Note the room temperature (t) and the barometric pressure in mm. Represent this difference by the symbol P.

4.4.4.2.4 Calculation of liberated gas volume - Calculate the volume of gas in ml, at standard conditions, liberated in the test described in 4.4.4.2.3, using the values described by the symbols in 4.4.4.2.2 and 4.4.4.2.3 in the following formula:

$$V = \frac{(A + C(B-H))\ 273P}{760(273 + t)} - \frac{(A + C(B_1 - H_1))\ 273P_1}{760(273 + t_1)}$$

4.4.4.3 Net increase in gas evolution due to the alpha-nitronaphthalene-explosive combination - The net increase in gas evolution due to the alpha-nitronaphthalene-explosive combination shall then be computed by the following formula:

$$V_{Net} = V_{R + A} - V_A - V_R :$$

Where:

$V_{R + A}$ = ml gas evolved per 2 grams of the alpha-nitronaphthalene-explosive combination sample

V_A = ml gas evolved per gram of the explosive sample

V_R = ml gas evolved per gram of the alpha-nitronaphthalene sample

V_{Net} = net increase in gas evolution due to the alpha-nitronaphthalene-explosive combination, ml per 2 grams

4.4.5 Rejection criteria - If a test specimen fails to meet any of the tests required by this specification, the lot represented by the sample shall be rejected.

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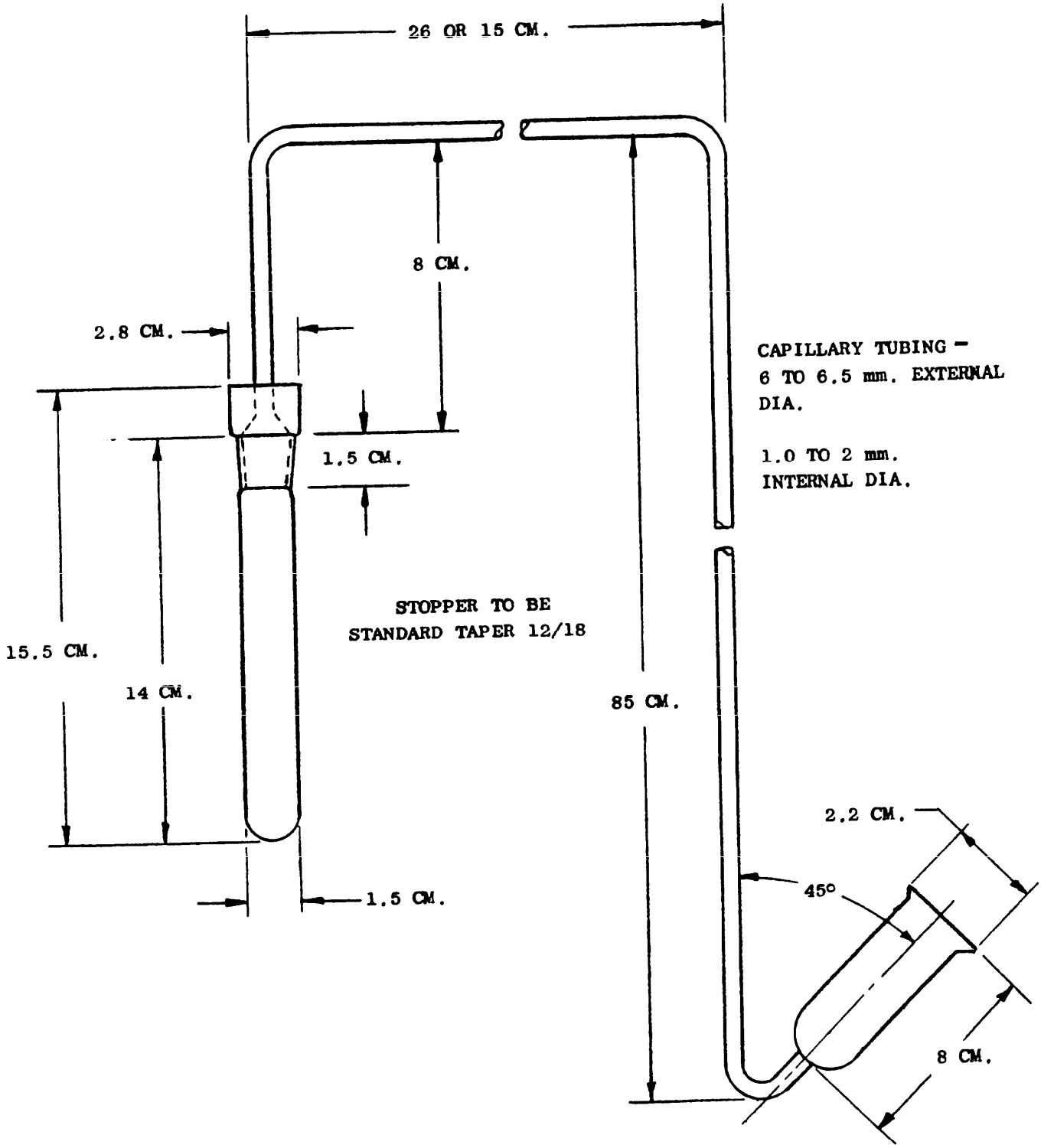


FIGURE 1: - APPARATUS FOR 100°C. VACUUM STABILITY TEST

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5. PREPARATION FOR DELIVERY

5.1 Preservation and packaging -

5.1.1 Level A - Not applicable.

5.1.2 Level B - Not applicable.

5.1.3 Level C -

5.1.3.1 Cleaning, drying and preservation application - Not applicable.

5.1.3.2 Unit packaging - Unless otherwise specified, unit packaging of the material shall be in clean, dry containers of the size and type specified by the purchase order.

5.2 Packing -

5.2.1 Level A - Not applicable.

5.2.2 Level B - Not applicable.

5.2.3 Level C - Material packaged in accordance with 5.1.3.2 shall be packed to afford protection against damage during shipment from the supply source to the first receiving activity for immediate use. Containers shall comply with Consolidated Freight Classification Rules or other common carrier regulations applicable to the mode of transportation.

5.3 Marking -

5.3.1 Special markings - Each container shall have the following special marking:

WARNING
HAZARDOUS SOLID
ABSORBED THROUGH SKIN

Avoid contact with skin, eyes and clothing.
Avoid breathing dust.
Use only with adequate ventilation.
In case of contract, immediately flush skin or eyes with plenty of water for at least 15 minutes; for eyes, get medical attention.
Remove and wash clothing before reuse.

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5.3.2 Normal markings - In addition to the markings required by contract or order, unit packages and shipping containers shall be marked in accordance with the requirements of MIL-STD-129.

6. NOTES

6.1 Intended use - Alpha-nitronaphthalene covered by this specification is intended for use in explosives.

6.2 Ordering data - Procurement documents should specify the following:

(a) Title, number and date of this specification.
(b) Quantity required in pounds.
(c) Applicable levels of packing and marking with requirements in detail, if other than as specified in Section 5.
(d) Size and type of container desired.

6.2.1 Criteria for use of proper level of preservation, packaging, and packing shall be as follows:

For Level A: This level shall be used for those items which are to be shipped to indeterminate destinations or stored under indeterminate conditions for redistribution anywhere.

For Level B: This level shall be used only when it is definitely known that the item will be held in covered storage in overseas locations for 6 months or less or in domestic locations for an indefinite period.

For Level C: This level shall be used only when it is definitely known that the packaged item is to be shipped to domestic installations for immediate use at the first receiving activity.

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SPECIFICATION ANALYSIS SHEET
NAVWEPS FORM 4121/3 (8-62)

FORM APPROVED
BUDGET BUREAU NO. 45-R309

INSTRUCTIONS

This sheet is to be filled out by Government or contractor personnel involved in the use of this specification in procurement of products for ultimate use by the activity shown on the reverse of this sheet.

This sheet is provided for obtaining information which will aid the activity shown in insuring that through use of this specification suitable products can be procured with a minimum amount of delay and at the least cost.

Comments and the return of this sheet will be appreciated.

Fold on dotted lines on reverse side, staple in corner, and send to the activity shown.

SPECIFICATION *(No. and abbrev. title)*

ORGANIZATION	CITY	STATE

CONTRACT NUMBER	QUANTITY OF ITEMS PROCURED	DOLLAR AMOUNT
		$

MATERIAL PROCURED UNDER A DIRECT GOVERNMENT CONTRACT ☐ OR A SUBCONTRACT ☐

1. HAS ANY PART OF THIS SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?
(If so, give paragraph number and wording, and recommendations for correcting the deficiencies.)

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?
☐ YES ☐ NO IF YES, IN WHAT WAY?

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification. If additional papers are attached, send this form and papers in an envelope. This form is addressed for use in window envelope when appropriate.)*

SUBMITTED BY *(Print name and title)*	DATE

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BUREAU OF NAVAL WEAPONS
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